

Creative Advertising Multi-Agent System

OpenAI Agents SDK Pipeline:
Creative Director → Strategist → Copywriter

Submitted by:	Alok Mehta
Assignment:	Part B - Assignment 2B
Course:	AI Course
Date:	May 12, 2026
Test Brief:	Launch a campaign for a new eco-friendly water bottle in Bali.

1. One-Page Explanation

1.1 Role of Each Agent

Creative Director

Receives the raw campaign brief and produces 3 - 5 fully formed campaign ideas. Each idea includes a campaign name, a one-line tagline, and a 2 - 3 sentence concept description. The agent's instructions push it toward boldness and cultural specificity, so ideas are tailored to the named market (in this case, Bali) rather than generic global copy.

Strategist

Takes the Creative Director's raw ideas and applies analytical judgment. It evaluates every idea against three criteria - market fit, originality, and potential reach - then selects the **top two** and articulates the reasoning for each selection. This gate prevents weak concepts from moving forward and ensures the copy stage focuses creative energy on the strongest strategic options.

Copywriter

Receives only the two approved campaigns and produces two tweets per campaign (four total). The agent is constrained to a 280-character limit per tweet, 2 - 3 hashtags, and a defined brand voice (eco-conscious, aspirational, travel-forward). Operating at the end of the pipeline, it can write with confidence because creative and strategic decisions are already resolved.

1.2 Tools and Functions Used

Component	Description
openai-agents SDK	Official OpenAI multi-agent framework (pip install openai-agents)
Agent class	Defines each agent with a name, model, and instructions string
Runner.run()	Async function that executes an agent given a string input
result.final_output	String property that holds the agent's completed response
asyncio.run()	Python standard library; runs the async pipeline from a sync entry point

No custom function-calling tools were added; all three agents operate purely on text instructions, which is sufficient for this ideation-and-copy workflow. Tool augmentation (for example, a web-search tool for trend data) could be added to the Strategist in a production version.

1.3 Why a Multi-Agent Approach Improves the Workflow

- **Separation of concerns.** A single all-in-one prompt for 'generate ideas, pick the best two, and write tweets' produces mediocre results because the model must simultaneously wear three hats. Splitting responsibilities lets each agent be optimized with a tightly scoped instruction set that matches one cognitive mode: divergent (Creative Director), evaluative (Strategist), and executorial

(Copywriter).

- **Quality gates.** The Strategist acts as a checkpoint. Only campaigns that survive strategic scrutiny reach the Copywriter, which prevents wasted copy work on ideas that would never run.
- **Auditability.** Because each stage produces a distinct, readable output, a human reviewer can intervene between stages - approve the ideas before strategy, approve strategy before copy - exactly as a real agency workflow would function.
- **Scalability.** Additional agents (a Legal Reviewer, a Visual Concept Agent, a Budget Estimator) can be inserted into the pipeline without rewriting existing agents, making the architecture extensible.

2. Implementation (creative_advertising_pipeline.py)

The full source for the multi-agent pipeline. The three agents are defined with the *OpenAI Agents SDK*; the **run_pipeline()** coroutine wires them in sequence so each agent's **final_output** becomes input to the next.

```
"""
Creative Advertising Multi-Agent System
Assignment: Part B - OpenAI Agents SDK Pipeline
Author: RetirementWorks / Alok Mehta

Pipeline: Creative Director → Strategist → Copywriter

Install dependencies:
    pip install openai-agents

Set your API key:
    set OPENAI_API_KEY=your-key-here (Windows)
    export OPENAI_API_KEY=your-key-here (Mac/Linux)
"""

import asyncio
import os
import sys
from agents import Agent, Runner

sys.stdout.reconfigure(encoding="utf-8")

# -----
# Agent Definitions
# -----

creative_director = Agent(
    name="Creative Director",
    model="gpt-4o-mini",
    instructions="""
You are a Creative Director at a world-class advertising agency.

Given a campaign brief, generate exactly 3 to 5 distinct and creative campaign ideas.

For each idea provide:
- Campaign Name: a short, memorable title
- Tagline: one punchy sentence that captures the essence
- Concept: 2-3 sentences describing the core idea, visual theme, and target emotion

Number each idea clearly (1, 2, 3, ...). Be bold, original, and culturally relevant
to the specified market.
""",
)

strategist = Agent(
    name="Strategist",
    model="gpt-4o-mini",
    instructions="""
You are a Senior Marketing Strategist at a world-class advertising agency.

You will receive a list of campaign ideas from the Creative Director.
Your job is to:
1. Briefly evaluate each idea on three criteria: market fit, originality, and

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- potential reach/impact (1-2 sentences each).
- 2. Select the TOP TWO campaign ideas.
- 3. For each selected campaign explain WHY it was chosen - cover target audience alignment, competitive differentiation, and execution feasibility.

Format your response as:

EVALUATION

[numbered evaluations of all ideas]

SELECTED CAMPAIGNS

Campaign 1: [Name]

Reason: [explanation]

Campaign 2: [Name]

Reason: [explanation]

""",

)

```
copywriter = Agent(
    name="Copywriter",
    model="gpt-4o-mini",
    instructions=""
```

You are a Social Media Copywriter specializing in viral, purpose-driven content.

You will receive two selected campaign ideas. For EACH campaign write exactly 2 tweets that:

- Are under 280 characters each (strictly enforced)
- Include 2-3 relevant hashtags
- Open with a hook (question, bold claim, or vivid image)
- End with a clear call-to-action or emotional close
- Match an eco-conscious, aspirational, travel-forward brand voice

Format your response as:

--- [Campaign Name] ---

Tweet 1: [text]

Tweet 2: [text]

--- [Campaign Name] ---

Tweet 1: [text]

Tweet 2: [text]

""",

)

```
# -----
# Pipeline Runner
# -----
```

```
async def run_pipeline(campaign_brief: str) -> dict:
    """
    Execute the three-agent creative pipeline sequentially.
    Each agent's output feeds directly into the next agent's input.

    Returns a dict with the output from each stage.
    """

    separator = "=" * 65
    print(separator)
```

```

print("  CREATIVE ADVERTISING MULTI-AGENT PIPELINE")
print(separator)
print(f"\n  Brief: {campaign_brief}\n")

# -----
# Stage 1 - Creative Director generates campaign ideas
# -----
print("-" * 65)
print("  STAGE 1 | CREATIVE DIRECTOR - Generating Campaign Ideas")
print("-" * 65)

stage1_prompt = (
    f"Campaign brief: {campaign_brief}\n\n"
    "Generate 3-5 creative campaign ideas for this brief."
)
result1 = await Runner.run(creative_director, input=stage1_prompt)
ideas: str = result1.final_output
print(ideas)

# -----
# Stage 2 - Strategist selects the top two ideas
# -----
print("\n" + "-" * 65)
print("  STAGE 2 | STRATEGIST - Selecting the Top Two Campaigns")
print("-" * 65)

stage2_prompt = (
    "Here are the campaign ideas from our Creative Director:\n\n"
    f"{ideas}\n\n"
    "Evaluate all ideas and select the top two. Explain your reasoning."
)
result2 = await Runner.run(strategist, input=stage2_prompt)
selected: str = result2.final_output
print(selected)

# -----
# Stage 3 - Copywriter writes promotional tweets
# -----
print("\n" + "-" * 65)
print("  STAGE 3 | COPYWRITER - Writing Promotional Tweets")
print("-" * 65)

stage3_prompt = (
    "The Strategist has selected these two campaigns:\n\n"
    f"{selected}\n\n"
    "Write 2 tweets for each selected campaign."
)
result3 = await Runner.run(copywriter, input=stage3_prompt)
tweets: str = result3.final_output
print(tweets)

print("\n" + separator)
print("  PIPELINE COMPLETE")
print(separator)

return {
    "campaign_ideas": ideas,
    "selected_campaigns": selected,
    "tweets": tweets,
}

# -----
# Entry Point

```

```
# -----  
  
if __name__ == "__main__":  
    api_key = os.getenv("OPEN_AI_API")  
    if not api_key:  
        raise EnvironmentError(  
            "OPEN_AI_API environment variable is not set. "  
            "Run: set OPEN_AI_API=your-key-here"  
        )  
    # The SDK reads OPENAI_API_KEY internally; map our var to it  
    os.environ["OPENAI_API_KEY"] = api_key  
  
    brief = "Launch a campaign for a new eco-friendly water bottle in Bali."  
    asyncio.run(run_pipeline(brief))
```

3. Sample Output

Verbatim console output from a real run of the pipeline against the test brief: *"Launch a campaign for a new eco-friendly water bottle in Bali."* Content is from **results.txt**, decoded from UTF-16. Decorative emojis from the original tweets have been stripped because ReportLab's built-in fonts do not render emoji glyphs; all other text is unchanged.

```
=====
CREATIVE ADVERTISING MULTI-AGENT PIPELINE
=====

Brief: Launch a campaign for a new eco-friendly water bottle in Bali.

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STAGE 1 | CREATIVE DIRECTOR - Generating Campaign Ideas
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### 1. **Campaign Name: Bottled in Balance**
- **Tagline:** Embrace Nature, Sip Sustainably.
- **Concept:** This campaign will highlight the symbiotic relationship between the eco-friendly water b

### 2. **Campaign Name: Wave of Change**
- **Tagline:** Drink Deeply, Live Lightly.
- **Concept:** Using powerful imagery of ocean waves and the biodiversity of Bali's beaches, this campa

### 3. **Campaign Name: Refresh the Island**
- **Tagline:** Hydrate with Purpose, Love Your Home.
- **Concept:** This campaign promotes a community-driven initiative where each purchase supports local

### 4. **Campaign Name: Flow with Nature**
- **Tagline:** Stay Hydrated, Stay Conscious.
- **Concept:** This imaginative campaign will use flowing animations and vibrant colors that echo the s

### 5. **Campaign Name: The Green Sip**
- **Tagline:** One Sip, One Planet.
- **Concept:** This powerful campaign juxtaposes a simple act - drinking water - with monumental enviro

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STAGE 2 | STRATEGIST - Selecting the Top Two Campaigns
-----
EVALUATION
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1. **Bottled in Balance**
- **Market Fit:** This campaign targets eco-conscious consumers effectively, appealing to their desire
- **Originality:** The concept is visually stunning, but the theme of harmony with nature is a familiar
- **Potential Reach/Impact:** Strong potential as it taps into the emotional connection with Bali's lan

2. **Wave of Change**
- **Market Fit:** This idea aligns well with the increasing consumer awareness around plastic pollution
- **Originality:** The imagery of ocean waves linked with storytelling offers a fresh angle on environm
- **Potential Reach/Impact:** High impact potential, particularly with consumers who are ocean enthusia

3. **Refresh the Island**
- **Market Fit:** Perfect for consumers interested in community and conservation initiatives, enhancing
- **Originality:** While community-focused campaigns are becoming popular, this angle with real locals
- **Potential Reach/Impact:** Could create a strong local community impact, but may have less mass appe

4. **Flow with Nature**
```


- **Market Fit:** Appeals to a niche of aesthetic eco-conscious consumers but may lack broader market reach
 - **Originality:** Unique in its approach of using animations and vibrant visuals, but risks being too niche
 - **Potential Reach/Impact:** Potentially appealing to design lovers, but may struggle to connect emotionally with a wider audience
5. **The Green Sip**
- **Market Fit:** Effectively addresses the growing trend of individual responsibility in environmental action
 - **Originality:** Directly linking consumption to environmental action is compelling, though similar to other eco-brands
 - **Potential Reach/Impact:** Strong potential due to relatable and everyday scenarios that encourage sustainable choices

SELECTED CAMPAIGNS

Campaign 1: **Wave of Change**

Reason: This campaign has been chosen for its strong alignment with a growing demographic of consumers who value sustainability and social responsibility.

Campaign 2: **The Green Sip**

Reason: This campaign effectively resonates with a wide audience by linking the simple act of drinking water to environmental stewardship.

STAGE 3 | COPYWRITER - Writing Promotional Tweets

--- Wave of Change ---

Tweet 1: Imagine a world where every wave tells a story of conservation and hope! Join us in protecting our oceans.

Tweet 2: Each visit to the ocean is a reminder of its beauty... and its fragility. Are you ready to be part of the change?

--- The Green Sip ---

Tweet 1: Every sip you take can be a step towards a cleaner planet! Discover how a simple glass of water can make a difference.

Tweet 2: Water shouldn't just quench your thirst; it can heal our planet! Join the Green Sip movement and make a difference.

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PIPELINE COMPLETE

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